

vides enough air pressure for the astronaut to survive the return to Earth during an emergency landing. Should an astronaut be forced to bail out over cold water, the pressurised suit would also provide thermal protection.

The suit provides an emergency oxygen system; parachute harness; parachute pack with automatic opener, pilot chute, drogue chute and main canopy; a life raft; two litres of emergency drinking water; flotation devices; and survival vest pockets containing a radio/beacon, signal mirror, shroud cutter, pen gun flare kit, sea dye marker, smoke flare and beacon. The attached parachute can be opened automatically or manually.



The Launch and Entry Suit

Space Suits

Given that the first spacewalk happened way back in 1965 and man landed on the moon in 1969, it is to be expected that the spacesuit has undergone



The Extravehicular Mobility Unit spacesuit being used on a spacewalk

some radical changes since then. To go through the history of each spacesuit wouldn't serve much purpose. There are three modern variants used today, the Russian Orlan space suit, the Chinese Feitian suit and the most advanced one – the American Extra-vehicular Mobility Unit (EMU).

The EMU is like a personal mini-spacecraft. The complex garment not only protects from the extreme conditions of space, it is in itself a mobile life support system with an oxygen supply, electrical power, water-cooling equipment, ventilating fan, and an in-suit drink bag.

► **Primary Life Support Subsystem (PLSS)** – The PLSS is